

Follow the Gradient

The power of differentiation

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- The big idea: optimisation by following gradients
- Recap: what are gradients and how do we find them?
- Jacobians as local linear maps
- Jacobian-vector and vector-Jacobian products

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- Fundamentally, we're interested in machines that we train by optimising parameters
 - How do we select those parameters?
- In deep learning/differentiable programming we typically define an objective function that we *minimise* (or *maximise*) with respect to those parameters
 - This implies that we're looking for points at which the gradient of the objective function is zero w.r.t the parameters

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The big idea: optimisation by following gradients

- Gradient based optimisation is a *big* field!
 - First order methods, second order methods, subgradient methods...
- With deep learning we're primarily interested in first-order methods¹.
 - Primarily using variants of gradient descent: a function $F(\mathbf{x})$ has a minima² (or a saddle-point) at a point $\mathbf{x} = \mathbf{a}$ where $\mathbf{a}$ is given by applying $\mathbf{a}_{n+1} = \mathbf{a}_n - \alpha \nabla F(\mathbf{a}_n)$ until convergence from some initial point $\mathbf{a}_0$.

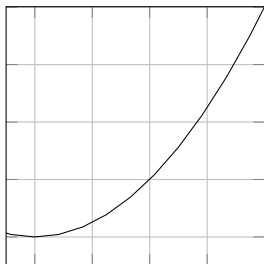
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²not necessarily global or unique

Recap: what are gradients and how do we find them?

The derivative in 1D

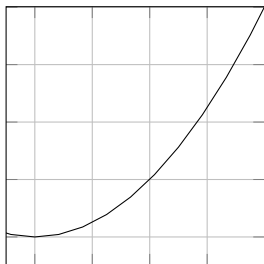
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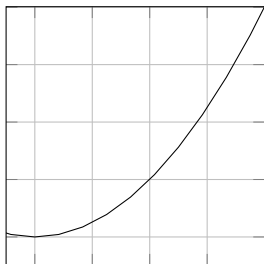
- Recall that the gradient of a straight line is $\frac{\Delta y}{\Delta x}$.
- For an arbitrary real-valued function, $f(a)$, we can approximate the derivative, $f'(a)$ using the gradient of the *secant line* defined by $(a, f(a))$ and a point a small distance, h , away $(a + h, f(a + h))$: $f'(a) \approx \frac{f(a+h) - f(a)}{h}$.



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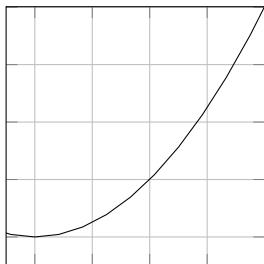
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- This expression is 'Newton's Quotient' or 'Fermat's Difference Quotient'.



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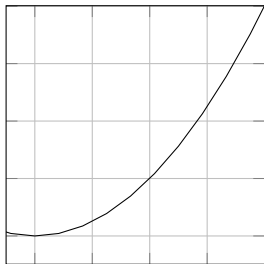


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- This expression is 'Newton's Quotient' or 'Fermat's Difference Quotient'.
- As h becomes smaller, the approximated derivative becomes more accurate.
- If we take the limit as $h \rightarrow 0$, then we have an exact expression for the derivative:
$$\frac{df}{da} = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}.$$



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The derivative of $y = x^2$ from first principles

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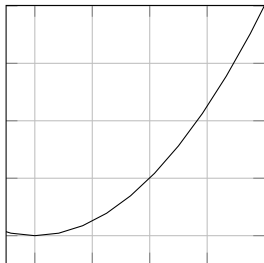
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Aside: numerical approximation of the derivative

- For numerical computation of derivatives it is better to use a “centralised” definition of the derivative:

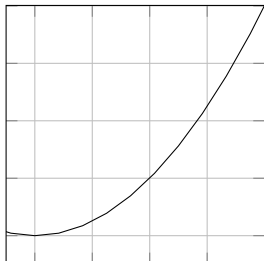
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 - The bit inside the limit is known as the *symmetric difference quotient*

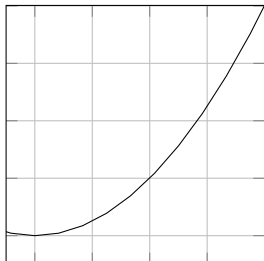


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- The bit inside the limit is known as the *symmetric difference quotient*
- For small values of h this has less error than the standard one-sided difference quotient.



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Aside: numerical approximation of the derivative

- If you are going to use difference quotients to estimate derivatives you need to be aware of potential rounding errors due to floating point representations.
 - Calculating derivatives this way using less than 64-bit precision is rarely going to be useful. (Numbers are not represented exactly, so even if h is represented exactly, $x + h$ will probably not be)
 - You need to pick an appropriate h - too small and the subtraction will have a large rounding error!

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Derivatives of deeper functions

- Deep learning is all about optimising deeper functions; functions that are compositions of other functions
 - e.g. $z = f \circ g(x) = f(g(x))$

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- Deep learning is all about optimising deeper functions; functions that are compositions of other functions
 - e.g. $z = f \circ g(x) = f(g(x))$
- The chain rule of calculus tells us how to differentiate compositions of functions:
 - $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$

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Example: differentiating $z = x^4$

Note that this is a silly example that just serves to demonstrate the principle!

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Equivalently, from first principles:

$$z = x^4$$

$$\frac{dz}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h}$$

$$\frac{dz}{dx} = \lim_{h \rightarrow 0} \frac{h^4 + 4h^3x + 6h^2x^2 + 4hx^3 + x^4 - x^4}{h}$$

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 - Thus the derivative is a vector (the 'tangent vector'),
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 - Thus the derivative is a vector (the 'tangent vector'),
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 - Equivalently, $\mathbf{y}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{y}(t+h) - \mathbf{y}(t)}{h}$ if the limit exists.

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Functions of multiple variables: partial differentiation

- What if the function we're trying to deal with has multiple variables³ (e.g. $f(x, y) = x^2 + xy + y^2$)?
 - This expression has a pair of *partial derivatives*, $\frac{\partial f}{\partial x} = 2x + y$ and $\frac{\partial f}{\partial y} = x + 2y$, computed by differentiating with respect to each variable x and y whilst holding the other(s) constant.

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$$\frac{\partial f}{\partial x_i}(a_1, \dots, a_n) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h}.$$

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- The vector of partial derivatives of a scalar-value multivariate function, $f(x_1, \dots, x_n)$ at a point $(a_1, \dots, a_n)$, can be arranged into a vector: $\nabla f(a_1, \dots, a_n) = \left(\frac{\partial f}{\partial x_1}(a_1, \dots, a_n), \dots, \frac{\partial f}{\partial x_n}(a_1, \dots, a_n) \right)^\top$.

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- In the case of a vector-valued multivariate function, the partial derivatives form a matrix called the **Jacobian**.

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Functions of vectors and matrices: partial differentiation

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 - They involve multiple variables, which are often wrapped up in the form of vectors or matrices, and more generally tensors.
 - **How will we find the gradients of these?**

Recap: what are gradients and how do we find them?

The chain rule for vectors

Suppose that $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$, g maps from $\mathbb{R}^m$ to $\mathbb{R}^n$ and f maps from $\mathbb{R}^n$ to $\mathbb{R}$.

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If $\mathbf{y} = g(\mathbf{x})$ and $z = f(\mathbf{y})$, then

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$$\frac{\partial z}{\partial x_i} = \sum_j \frac{\partial z}{\partial y_j} \frac{\partial y_j}{\partial x_i}.$$

Equivalently, in vector notation:

$$\nabla_{\mathbf{x}} z = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^\top \nabla_{\mathbf{y}} z$$

where $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$ is the $n \times m$ Jacobian matrix of g .

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{XW})$

- Let $\mathbf{D} = \mathbf{XW}$ where the rows of $\mathbf{X} \in \mathbb{R}^{n \times m}$ contain some fixed *features*, and $\mathbf{W} \in \mathbb{R}^{m \times h}$ is a matrix of weights.
- Also let $\mathcal{L} = f(\mathbf{D})$ be some scalar function of $\mathbf{D}$ that we wish to minimise.

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- Also let $\mathcal{L} = f(\mathbf{D})$ be some scalar function of $\mathbf{D}$ that we wish to minimise.
- What are the derivatives of $\mathcal{L}$ with respect to the weights $\mathbf{W}$?

Recap: what are gradients and how do we find them?

Example: $\nabla_w f(\mathbf{XW})$

- Start by considering a specific weight, W_{uv} : $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}}$.

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- Therefore, we can simplify the summation to only consider cases where $j = v$: $\sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}} = \sum_i \frac{\partial \mathcal{L}}{\partial D_{iv}} \frac{\partial D_{iv}}{\partial W_{uv}}$.

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- What is $\frac{\partial D_{iv}}{\partial W_{uv}}$?

$$D_{iv} = \sum_{k=1}^m X_{ik} W_{kv}$$

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Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Start by considering a specific weight, W_{uv} : $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}}$.
- We know that $\frac{\partial D_{ij}}{\partial W_{uv}} = 0$ if $j \neq v$ because D_{ij} is the dot product of row i of $\mathbf{X}$ and column j of $\mathbf{W}$.
- Therefore, we can simplify the summation to only consider cases where $j = v$: $\sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}} = \sum_i \frac{\partial \mathcal{L}}{\partial D_{iv}} \frac{\partial D_{iv}}{\partial W_{uv}}$.
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$$\therefore \frac{\partial D_{iv}}{\partial W_{uv}} = X_{iu}$$

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- and note that the sum over i is doing a dot product with row u and column v if we transpose X_{iu} to $X_{ui}^\top$: $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i X_{ui}^\top \frac{\partial \mathcal{L}}{\partial D_{iv}}$.

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- We can then see that if we want this for all values of $\mathbf{W}$ it simply generalises to: $\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{X}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{D}}$.

Recap: what are gradients and how do we find them?

That worked, but the indices were a pain!

- What did we actually do in the previous derivation?
 - We worked out which elements of $\mathbf{D}$ depended on a particular element W_{uv} .
 - We multiplied each local derivative by the corresponding sensitivity $\frac{\partial \mathcal{L}}{\partial D_{ij}}$.
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- This is all perfectly manageable for $\mathbf{D} = \mathbf{XW}$...
 - ...but we don't want to repeat that index-heavy argument for every operation in a deep model!

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- This is of course totally true...
- But, it isn't a particularly useful way to think about the gradients of a loss with respect to the weights of a parameterised function.
 - **The gradient of the loss with respect to a parameter tells you how much the loss will change with a small perturbation to that parameter.**

Jacobians as local linear maps

Packaging up all of those partial derivatives

- For a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$, the Jacobian collects all of the partial derivatives into an $n \times m$ matrix:

$$(J_f(\mathbf{x}))_{ji} = \frac{\partial f_j}{\partial x_i}.$$

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- More usefully, the Jacobian describes the best linear approximation to f close to $\mathbf{x}$ (c.f. the first-order Taylor Series expansion):

$$f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + J_f(\mathbf{x})\Delta\mathbf{x}.$$

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- So, the Jacobian tells us the first-order output change produced by a small input perturbation.

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 - ① Given a particular perturbation of the inputs, what happens to the outputs?
 - ② Given a particular sensitivity at the outputs, what are the corresponding sensitivities at the inputs?
- These are *Jacobian-vector products* and *vector-Jacobian products*.

Jacobian-vector products

Push a perturbation forwards

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- The corresponding first-order change in the output is the **Jacobian-vector product** (JVP):

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 - We can compute this product without ever constructing the complete Jacobian.

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- Equivalently, this is the gradient of the scalar-valued function $\mathbf{u}^\top f(\mathbf{x})$:

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- A VJP takes sensitivities at the output and propagates them back to the input.

JVPs and VJPs

Returning to $D = \mathbf{XW}$

- Consider the function $g(\mathbf{W}) = \mathbf{XW}$.

JVPs and VJPs

Returning to $\mathbf{D} = \mathbf{X}\mathbf{W}$

- Consider the function $g(\mathbf{W}) = \mathbf{X}\mathbf{W}$.
- If we perturb the weights by $\Delta\mathbf{W}$, the JVP pushes that perturbation forwards:

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- If $\mathbf{G} = \frac{\partial\mathcal{L}}{\partial\mathbf{D}}$ contains the sensitivities arriving from later in the computation, the VJP pulls them backwards:

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- This is exactly the result we obtained from the element-wise derivation.

JVPs and VJPs

The chain rule is composition of products

- If $h = g \circ f$, then a perturbation can be pushed forwards one operation at a time:

$$J_h(\mathbf{x})\mathbf{v} = J_g(f(\mathbf{x}))(J_f(\mathbf{x})\mathbf{v}).$$

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The chain rule is composition of products

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$$J_h(\mathbf{x})\mathbf{v} = J_g(f(\mathbf{x}))(J_f(\mathbf{x})\mathbf{v}).$$

- Likewise, a sensitivity can be pulled backwards one operation at a time:

$$J_h(\mathbf{x})^\top \mathbf{u} = J_f(\mathbf{x})^\top (J_g(f(\mathbf{x}))^\top \mathbf{u}).$$

JVPs and VJPs

What about matrices and higher-order tensors?

- Conceptually, we could flatten every tensor into a vector, construct the corresponding Jacobian, perform the product and reshape the result.
- In practice, we preserve the natural tensor shapes and directly compute the required product.
 - We avoid explicitly constructing a potentially enormous Jacobian.
 - The result has the natural shape of the input for a VJP, or the output for a JVP.
- Next week, we'll see how forward- and reverse-mode automatic differentiation compute these products through an arbitrary program.